

Enhancing Implicit Text Classification via Semi-Supervised Learning with GAN Variants

Abstract

Implicit sentences for text classification pose a significant challenge, as such sentences do not directly convey the intended meaning. Transformer-based architectures, such as BERT and RoBERTa, have demonstrated strong performance in text classification tasks. However, these models often require large amounts of labeled data and tend to struggle with capturing latent semantic structures, especially in the case of implicit content. Given the limited availability of labeled implicit sentences in most datasets, semi-supervised learning emerges as a promising alternative. Generative Adversarial Networks (GANs) have recently gained attention in this context. In this work, we investigate the effectiveness of two semi-supervised GAN-based models—GAN-BERT and Triple-GAN—for handling implicit classification under limited supervision. In this work, we demonstrate significant performance improvements of Triple-GAN over baseline models on two datasets, SMP2019-ECISA and Implicit Misogyny Classification (IMC). Using BERT as the baseline, Triple-GAN yields accuracy gains of 2.87% on SMP2019-ECISA and 3.22% on IMC. With RoBERTa as the baseline, improvements of 2.0% and 2.77% in accuracy are observed on the respective datasets. In terms of F1-score, Triple-GAN surpasses BERT by 2.81% on SMP2019-ECISA and 44.78% on IMC, also outperforming RoBERTa with F1-score gains of 2.2% and 30.6% on SMP2019-ECISA and IMC respectively. GAN-BERT also demonstrates a percentage increase on accuracy and F1-score on both datasets. These findings highlight the potential of generative models in capturing subtle semantic patterns and handling data-scarce scenarios in implicit text classification. Implementation details are available at: <https://github.com/kanduladeekshithareddy/Implicit-text-classification.git>

Keywords: Implicit sentiment analysis, implicit classification, Text classification, Natural Language processing

1 Introduction

Many text classification tasks are facing significant problem with the implicit sentences present in the dataset for instance, sentiment analysis [1–3], hate speech detection [4] and discourse relation classification [5]. Implicit Classification (IC) is categorizing the text based on implicit cues and inferred meaning in the sentence. For instance, if you consider an implicit sentence “There aren’t many things that are more satisfying than

telling a girl, ‘No.’” this clearly shows the superiority or empowerment over women but it is an implicit inferred sentence representing it as misogynistic sentence. IC requires model to capture the implicit cues that may be context dependent, inferred, sarcasm or irony.

Previously, to account for implicit cues, Bidirectional Encoder Representations from Transformers (BERT) [6] like pretrained transformer is used and is finetuned under the intended classification task focusing on the contextual understanding of the sentence. As the sentences lack explicitly stated words and are to be inferred, the contextual sentences around them are given importance, [1] worked on Context BERT through the query attention mechanism but it needs a high-quality dataset with context around the query sentence. For efficient feature extraction of text, [2, 3, 7] used supervised contrastive learning to capture the semantic representation of the sentences. To give more weight to the implicit cues and add additional information to account for explicit words, external information is taken into consideration like the use of parts of speech [8], word definitions from Wikipedia [9], these may add noise to the data already existing leading to inaccurate results. Importantly, most of these models depend on datasets rich in implicit content. In reality, the percentage of implicit sentences present in datasets is remarkably low, making supervised training insufficient.

To address this issue of data scarcity and improve classification performance, we propose a careful study of semi-supervised learning frameworks based on Generative Adversarial Networks (GANs). Specifically, we evaluate two architectures: GAN-BERT and Triple-GAN. GAN-BERT augments the standard BERT model with a discriminator that differentiates between real (labeled and unlabeled) and generated embeddings, encouraging better feature learning from unlabeled data. On the other hand, Triple-GAN introduces a third component—a classifier—which enables the generation of pseudo-labels and encourages consistency between data, labels, and feature representations. The major contributions of the proposed work include the following.

1. We explored GAN-BERT and Triple-GAN with limited supervision, to address the problem of data scarcity.
2. We adapt the Triple-GAN architecture to the domain of text classification, introducing a novel application of this framework.
3. Our experimental analysis highlights that Triple-GAN outperforms GAN-BERT and BERT in accuracy and F1-score with minimal supervision.

2 Literature Review

Traditional text classification often struggled with implicit expressions - those where opinions or emotions are conveyed indirectly through nuanced language. To address these challenges, contemporary studies have leveraged deep learning architectures combined with specialized mechanisms for capturing implicit cues. From basic Recurrent Neural Network (RNN) architectures like Long -Short Term Memory (LSTM), Gated Recurrent Unit (GRU) to complex attention models, transformers have been applied for identifying and weighting relevant contextual cues that indicate implicit sentences.

2.1 Implicit classification

[5] proposed a model that uses context – aware filters in convolution operation for implicit discourse relation classification but struggles with long-range dependencies. Focusing on sentence semantic and syntactic understanding,[10] proposed Bi-directional LSTM (BiLSTM) with multi-polar orthogonal attention to account for implicit emotion analysis. Multi-polar attention can identify subtle cues in the sentence by giving attention to the words that account for sentiment. Using query attention mechanism along with the contextual sentences around and pretrained language model [1] proposed context-BERT, in order to work on implicit sentiment analysis. [2] used supervised contrastive pre-training to analyze the distribution of sentiment labels in both implicit and explicit phrases to account for sentiment analysis. [11] addressed the confounding effects in sentences, proposed a causal intervention model to account for the dependencies of confounders when determining the sentiment of a sentence. [3] used supervised contrastive training for both context and the sentence by running parallelly on bi-affine modules.[4] has used both causality and contrastive learning called the Causality-Guided Contrastive Learning (CCL) for addressing the problem of implicit hate detection. As implicit sentences are deprived of explicit words in sentiment analysis,[12] augmented explicit sentiment by creating them using T5 as a backbone language model to enhance the classification ability. Finally, all these methods involve contextual understanding, feature extraction or external knowledge integration to account for the implicitness in sentences.

2.2 Semi-Supervised GAN

Recent advancements in GAN [13] have shown promising results in text and image classification, particularly with low-resource settings and semi-supervised learning. [14] proposed semi-supervised learning in GANs, that is Semi-Supervised GAN (SS-GAN), for the discriminator to be modified as a classifier with k-classes along with one more class that represents real or fake data. Extending this,[15] proposed GAN-BERT by combining pre-trained model like BERT with SS-GAN to state that even with less amount of labeled data 1%, 5%, 10%, 20%, 40%, 50% the model gets higher accuracy compared to the BERT model on various classification tasks. Further, the versatility of GAN-BERT has been validated across multiple domains by replacing the base BERT encoder with domain-specific language models such as SciBERT [16] for scientific texts and BioBERT [17] for biomedical corpora. These variations maintain the robustness of GAN-BERT in semi-supervised settings highlighting the generalizability of the architecture across different language modeling contexts.

3 Preliminaries

3.1 GAN-BERT

GAN-BERT [15] combines the strengths of a pre-trained transformer-based model, BERT [6], with the generative capabilities of GAN [13]. This hybrid architecture has demonstrated impressive performance even in scenarios with extremely limited

labeled data—achieving competitive results with as few as 100–200 training examples on various benchmark datasets, including SST-5 and 20 Newsgroups.

GAN-BERT is structured similarly to a standard GAN, consisting of two primary components: a Generator (G) and a Discriminator (D). Input sentences are first passed through BERT to obtain contextualized embeddings. These embeddings, along with synthetic embeddings generated by the generator, are then fed into the discriminator. The discriminator is trained not only to distinguish between real and fake embeddings but also to classify real BERT embeddings into one of the k task-specific classes. Meanwhile, the synthetic embeddings generated by G are ideally classified by D into an additional $(k + 1)$ class, indicating fake or generated data. This structure enables the model to learn from both labeled and unlabeled data, improving generalization and robustness in low-resource settings.

3.2 Triple-GAN

Triple GAN [18] is used for image classification tasks earlier, observing its error rates percentage compared to other GAN variants, the model has achieved state of art results with very less amount of labeled data. Triple GAN contains three modules, classifier (C) which aims to consider unlabeled data to predict the class. Generator (G) generates class conditional data and the discriminator (D) determines real or fake data using both the sentence and the label information.

3.2.1 Three-player game

As said earlier, classifier predicts the class label y , given the input data x . Formally, $P_c(y | x)$ is characterized by the classifier C . Let $P_g(x | y)$ denote the class-conditional generation by the generator G . The joint distribution of real labeled data is denoted by $P(x, y)$. The discriminator D considers all joint distributions of pseudo-generated data, $P_g(x, y)$, and classifier data, $P_c(x, y)$, as fake samples, and the real labeled data $P(x, y)$ as real samples. The utility involved in the three-player game is defined as follows by [18]:

$$\begin{aligned} \min_{G, C} \max_D \mathcal{L}(D, G, C) = & \mathbb{E}_{(x, y) \sim P(x, y)} [\log D(x, y)] \\ & + \mathbb{E}_{(x, y) \sim P_g(x, y)} [\log(1 - D(x, y))] \\ & + \mathbb{E}_{(x, y) \sim P_c(x, y)} [\log(1 - D(x, y))] + R_L, \end{aligned} \quad (1)$$

where R_L denotes the Kullback–Leibler (KL) divergence between the classifier-induced distribution $P_c(x, y)$ and the real data distribution $P(x, y)$, i.e.,

$$R_L = \text{KL}(P_c(x, y) \parallel P(x, y))$$

4 Methodology

In our work, we utilized the standard GAN-BERT architecture as originally proposed, while the Triple-GAN framework is modified to suit the requirements of our task as

shown in figure 1. Each input batch includes both labeled and unlabeled sentences. The sentences are denoted as $X = \{x_1, x_2, \dots, x_n\} \in \mathbb{R}^{d1}$. Each labeled sentence x_i is associated with one of k class labels, i.e., $\{0, 1, \dots, k\}$, where the classes may represent categories such as neutral, positive, and negative for ISA, or misogynistic and non-misogynistic for IMC.

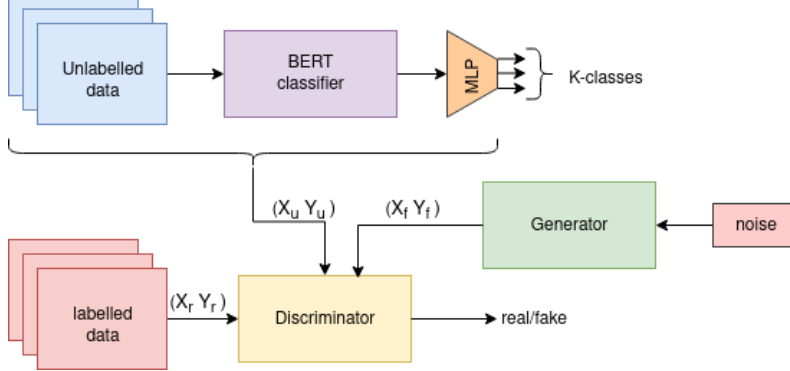


Fig. 1: Text based Triple-GAN Model

BERT [6] is a transformer-based architecture that is pre-trained for general tasks and can be fine-tuned for a particular downstream task. BERT is able to capture the contextual embeddings from the text provided, so it is used as a classifier C . We employ a BERT encoder followed by a Multi-Layer Perceptron (MLP). C receives an unlabeled sentence denoted as $X_u = \{x_{u1}, x_{u2}, \dots, x_{un}\} \in \mathbb{R}^{d2}$ and outputs a k -dimensional vector representing class probabilities. The predicted label Y_u is obtained as:

$$Y_u = \arg \max_k C(X_u). \quad (2)$$

The generator G is an MLP designed to generate data based on randomly generated labels. G takes as input a 100-dimensional noise vector $z \sim \mathcal{N}(\mu, \sigma^2)$ sampled from a Gaussian distribution and a class label $y_f \sim \mathcal{U}(\{1, 2, \dots, k\})$. It generates a synthetic sample $X_f \in \mathbb{R}^d$. Discriminator D is also an MLP designed to determine real or pseudo samples based on input and labels associated with it. D receives concatenated representations from C and G as input and produces a probability distribution over $k + 1$ classes: k corresponding to real labeled data and the $(k + 1)^{\text{th}}$ class representing synthetic samples produced by G . The loss functions are :

Generator Loss ($\mathcal{L}_G$) : The loss associated with G is the adversarial loss from D , which tries to distinguish between real and fake samples. G is trained to minimize this loss, thereby attempting to fool D , while D tries to maximize it:

$$\mathcal{L}_G = \mathbb{E}_{(x,y) \sim P_g(x,y)} [\log(1 - D(G(y, z), y))]. \quad (3)$$

¹ $\mathbb{R}^d$ indicates the vector space of sentence embeddings.

² $\mathbb{R}^d$ indicates the vector space of sentence embeddings.

Classifier Loss ($\mathcal{L}_C$) : The classifier C is optimized using three components:

1. *Adversarial Loss ($\mathcal{L}_C^{\text{adv}}$)*: This loss encourages C to fool the discriminator by producing samples that appear real:

$$\mathcal{L}_C^{\text{adv}} = \mathbb{E}_{(x,y) \sim P_c(x,y)} [P_c(y|x) \log(1 - D(x, y))]. \quad (4)$$

2. *Supervised Loss ($\mathcal{L}_{\text{RL}}$)*: This is the standard cross-entropy loss between the real label distribution and classifier predictions:

$$\mathcal{L}_{\text{RL}} = \mathbb{E}_{(x,y) \sim P(x,y)} [-\log P_c(y|x)]. \quad (5)$$

3. *Pseudo Discriminative Loss ($\mathcal{L}_{\text{Rp}}$)*: This term measures the divergence between the classifier’s predictions and generated pseudo-labels:

$$\mathcal{L}_{\text{Rp}} = \mathbb{E}_{(x,y) \sim P_g(x,y)} [-\log P_c(y|x)]. \quad (6)$$

The overall classifier loss is then computed as:

$$\mathcal{L}_C = \mathcal{L}_C^{\text{adv}} + \mathcal{L}_{\text{RL}} + \alpha_p \mathcal{L}_{\text{Rp}}, \quad (7)$$

where α_p is a hyperparameter that controls the contribution of the pseudo-discriminative loss.

Discriminator Loss ($\mathcal{L}_D$) : The discriminator D is trained to distinguish real labeled data from both generated and classifier-induced pseudo data. It is optimized using adversarial losses from all three joint distributions: $P(x, y)$, $P_g(x, y)$, and $P_c(x, y)$. The corresponding loss function is:

$$\begin{aligned} \mathcal{L}_D = & -\mathbb{E}_{(x,y) \sim P(x,y)} [\log D(x, y)] \\ & -\mathbb{E}_{(x,y) \sim P_g(x,y)} [\log(1 - D(x, y))] \\ & -\mathbb{E}_{(x,y) \sim P_c(x,y)} [\log(1 - D(x, y))]. \end{aligned} \quad (8)$$

where α determines the relative importance given to G and C , with $\alpha \in [0, 1]$. The training process follows the utility function as described in section 3.2.1, in order to optimize the losses associated with C , G , and D . In the testing phase, only the classifier C is used.

5 Experimental settings

5.1 Dataset

Our experiment is conducted on two datasets, one is for implicit sentiment classification (ISC) task, a Chinese dataset, SMP2019-ECISA³ and for implicit misogyny classification, IMC dataset. SMP2019-ECISA is collected from Weibo, a Chinese platform which enables people communication online. The dataset contains both labeled

³<https://www.biendata.xyz/competition/smpecisa2019/>

sentences along with the contextual sentences around. This dataset is processed in such a way that the sentences are filtered with explicit sentiment words and are labeled positive, negative and neutral. But for the experiment we only use the labeled target sentence out of which only 15% of them are implicit in nature. The statistics of this dataset are shown in table 1.

Table 1: [3] Statistics of SMP2019-ECISA dataset.

Dataset	Positive	Negative	Neutral	Total
Train	3828	3957	7003	14,788
Validation	1232	1358	2553	5143
Test	919	979	1902	3800

The other dataset is for misogyny classification task.[19] annotated data collected from reddit online platform. It contains about 6,567 sentences labeled as misogynistic and non-misogynistic. Later, the misogynistic content is again human annotated with implicit or explicit labels. Out of 570 misogynistic labels in dataset 221 are implicit in nature which accounts for 38% of data. The statistics are shown in table 2.

Table 2: Statistics of IMC dataset.

Dataset	Misogynistic	Non-Misogynistic	Total
Train	570	4693	5263
Test	129	1173	1302

5.2 Implementation Details and Evaluation Metrics

The Hugging-Face Transformers library in PyTorch’s pre-trained `hfl / Chinese-bert-wwm-ext` model⁴ is used as the variant of the BERT model in figure 1 for ISA task and `google-bert / bert-base uncased`⁵ model for IMC. 32 is the batch size that has been set for implicit sentiment analysis and 16 for implicit misogyny classification. As multi-layer perceptrons (MLPs), the discriminator (D) and generator (G) each have two hidden layers with Leaky ReLU activation function. Better generalization is ensured by setting the dropout rate at 0.5. The learning rates of classifier, generator and discriminator are set to 5×10^{-6} , 6×10^{-5} and 3×10^{-7} respectively. Kaggle environment with GPU Tesla P100-PCIE-16GB is used to run the model.

The performance of the model is evaluated using two primary metrics: **accuracy** and **F1-score**. Accuracy provides a general indication of how often the model makes correct predictions. It is calculated based on the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) using the following formula:

⁴<https://huggingface.co/hfl/chinese-bert-wwm-ext>

⁵<https://huggingface.co/google-bert/bert-base-uncased>

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (9)$$

F1-score is also used, as it offers a better balance between *precision* and *recall*. Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

Recall measures the proportion of correctly predicted positive instances among all actual positive instances:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

The F1-score is the harmonic mean of precision and recall, and is computed as:

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \times 100 \quad (12)$$

Together, these metrics provide a comprehensive evaluation of the model’s classification performance, particularly when dealing with datasets that have uneven class distributions.

6 Analysis and Results

The results for the SMP2019-ECISA dataset are presented in Table 3, while those for the IMC dataset are shown in Table 4. On IMC dataset, with 50% labeled data, Triple-GAN outperforms BERT, achieving an F1-score improvement of approximately 9.6%, and a gain of 2.8% on the SMP2019-ECISA dataset. In terms of accuracy, Triple-GAN shows an increase of 2.5% on the ISA task and 1.9% on the IMC task.

GAN-BERT also surpasses BERT across both datasets but with comparatively smaller margins. On the ISA task, it achieves a 2.0% improvement in accuracy and 1.8% in F1-score. On the IMC task, it records a 0.4% higher accuracy and a 4.1% increase in F1-score. These results highlight the superior performance of Triple-GAN in capturing implicit cues under limited supervision.

Table 3: Model Performance on SMP2019-ECISA.

Model	Labeled Data (%)	Accuracy (%)	Macro-F1 (%)
BERT	100	79.2	77.3
BERT	50	78.2	76.5
RoBERTa(base)	100	79.3	78.0
GAN-BERT	50	79.8	77.9
Triple-GAN	50	80.2	78.7

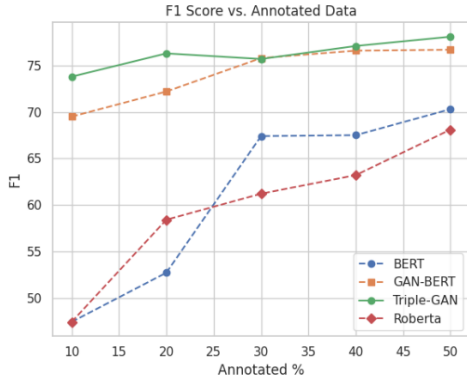
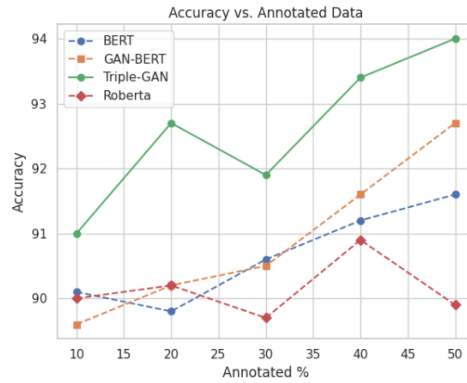
This is because GAN-BERT has a discriminator that is responsible for two tasks 1) to distinguish the samples to be real or fake, which is in turn helpful in optimization

Table 4: Model Performance on implicit misogyny dataset

Model	Labeled Data (%)	Accuracy (%)	Macro-F1 (%)
BERT	100	91.6	74.0
BERT	50	91.6	70.3
RoBERTa(base)	100	92.0	68.2
GAN-BERT	50	92.0	73.2
Triple-GAN	50	93.4	77.1

of generator. 2) The actual classification task, that is a real labeled sentences need to be categorized into one of the k-classes. This makes the performance of discriminator diminish. So, Triple-GAN comes into play, which has three separate models to account for the above two tasks.

Triple-GAN is able to capture the implicit nature of the sentence more accurately compared to GAN-BERT and BERT. The Generator and classifier in a Triple-GAN are to be in equilibrium for proper functioning of the three-player game. The discriminator plays a key role in their optimization. Generator, being able to learn the true distribution of the data given the label information has been of great advantage in triple-GAN. The use of label information with the true input sentence along with the model accounting for the less % of labeled data, implicit sentences are taken care of by our model.

**Fig. 2:** Graph plot of F1-score with different percentage of labeled data on implicit misogyny classification.**Fig. 3:** Graph plot of accuracy with different percentage of labeled data on implicit misogyny classification.

As a semi-supervised learning approach, When compared with transformer based architectures, which need large amounts of data, in figures 2,4 it is evident that even with just 20% of labeled data our model has acquired 35% higher F1-score on IMC dataset and about 10% higher F1-score on SMP2019-ECISA. This has proven that zero-shot learning tasks are worth a shot to improve their results. As we can see there is an imbalance in the class count in IMC dataset, with traditional transformer-like

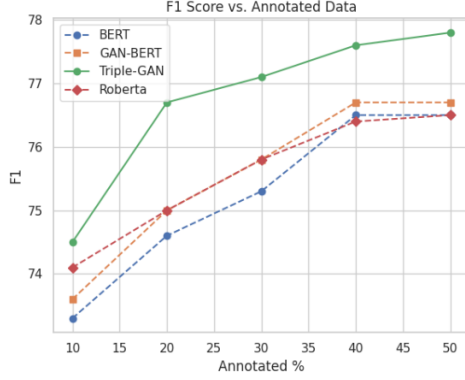


Fig. 4: Graph plot of F1-score with different percentage of labeled data on SMP2019-ECISA.

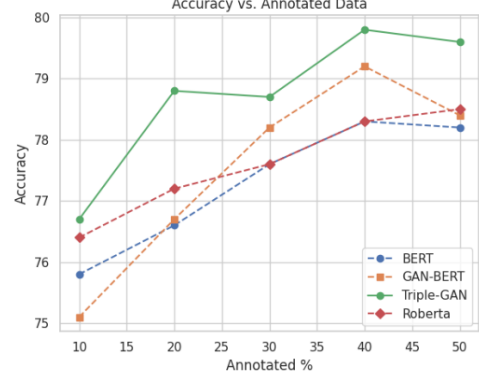


Fig. 5: Graph plot of accuracy with different percentage of labeled data on SMP2019-ECISA.

models, it may be possible that the model may be inclined to the class with a higher number of samples, but our model is not inclined to a particular class, as we can see that the F1-score is not too poor when compared to the accuracy, even when the data is too little, which is about 10%. When using 100% of the labeled data, both BERT and RoBERTa models did not achieve higher results compared to the Triple-GAN model trained with only 50% of the labeled data. Compared to Bert that uses 100% of labeled data, Triple-GAN acquired a gain of 1.8% in accuracy and 4.1% in F1-score on IMC dataset and a gain of 1.2% in accuracy and 1.8% in F1-score on SMP2019-ECISA. On the IMC dataset, the Triple-GAN model achieved a 1.9% increase in accuracy and a 4.1% increase in F1-score compared to RoBERTa. Similarly, on the ISA dataset, the Triple-GAN model showed a 1.1% increase in accuracy and a 0.8% increase in F1-score over RoBERTa.

6.1 Explainability of Triple-GAN

To further analyze the interpretability of our models, we employed LIME (Local Interpretable Model-agnostic Explanations)[20] to visualize the feature importance for individual predictions made by BERT and Triple-GAN on implicit misogyny sentences. The results are shown in Figure 6. For instance the sentence "Women are already equal now — there's no need for feminism anymore", It is an implicit sentence. Triple-GAN focuses on the latent meaning, correctly identifying the misogynistic undertone of the sentence despite the surface-level politeness. BERT, on the other hand, highlights words like "women" and "equal", but places less emphasis on the critical word "anymore". It struggles to capture the full sentiment and inadvertently misclassifies the sentence as misogynistic or not misogynistic, missing the deeper context that Triple-GAN successfully identifies.

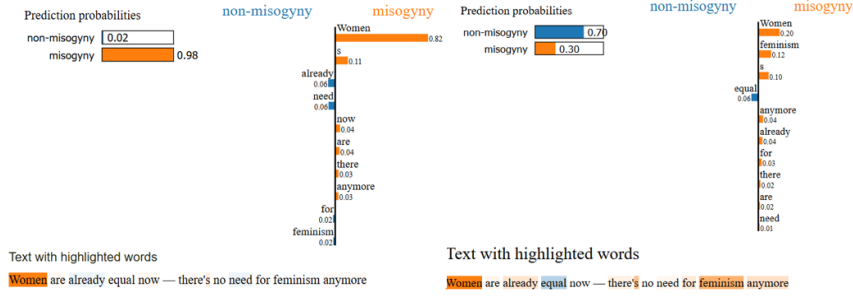


Fig. 6: LIME-based explanation for the sentence “Women are already equal now — there’s no need for feminism anymore”. Comparison between BERT (right) and Triple-GAN(left). The bars represent the contribution of each word to the predicted class, with red bars indicating words that increase the probability of detecting implicit misogyny.

7 Conclusion

In this work, we investigated the performance of Triple-GAN and GAN-BERT for implicit text classification within a semi-supervised learning framework. Unlike standard discriminative models such as BERT and RoBERTa, GAN-based variants incorporate a generative component, enabling the extraction of richer latent features and allowing robust performance even with limited annotated data.

Our experimental results demonstrate that GAN-based models, particularly Triple-GAN, outperformed strong transformer-based baselines. Notably, Triple-GAN achieved superior performance over RoBERTa and BERT even when trained with only 50% of labeled data. It could also take care of imbalance in the dataset, highlighting its effectiveness in low-resource and imbalanced dataset scenarios. Hence, we can say that Triple-GAN enhances classification performance under data-scarce and imbalance in dataset settings. Nevertheless, our findings suggest that GAN-based approaches present a promising supplement to transformer-based classifiers, particularly for low-data classification tasks. Future work could focus extending the framework to multilingual and cross-domain settings, and exploring hybrid GAN-transformer models, could further enhance performance due to generalization.

References

- [1] Yin, W., Shang, L.: Contextbert: Enhanced implicit sentiment analysis using implicit-sentiment-query attention. In: 2022 International Joint Conference on Neural Networks (IJCNN), pp. 1–8 (2022). IEEE
- [2] Li, Z., Zou, Y., Zhang, C., Zhang, Q., Wei, Z.: Learning implicit sentiment in aspect-based sentiment analysis with supervised contrastive pre-training. arXiv preprint arXiv:2111.02194 (2021)

- [3] Kong, L., Yang, W., Wei, F.: Implicit emotion analysis based on improved supervised contrastive learning. *Electronics* **12**(10), 2172 (2023)
- [4] Jiang, T.: Learn from failure: Causality-guided contrastive learning for generalizable implicit hate speech detection. In: *Proceedings of the 31st International Conference on Computational Linguistics*, pp. 8858–8867 (2025)
- [5] Munir, K., Zhao, H., Li, Z.: Learning context-aware convolutional filters for implicit discourse relation classification. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **29**, 2421–2433 (2021)
- [6] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (long and Short Papers)*, pp. 4171–4186 (2019)
- [7] Fu, J., Li, X., Zhu, Y., Du, Y., Fan, Y., Chen, X., Huang, D., Wang, S.: An implicit aspect-based sentiment analysis method using supervised contrastive learning and knowledge embedding. *Applied Soft Computing* **167**, 112233 (2024)
- [8] Wang, J., Li, X., He, J., Zheng, Y., Ma, J.: Enhancing implicit sentiment learning via the incorporation of part-of-speech for aspect-based sentiment analysis. In: *China National Conference on Chinese Computational Linguistics*, pp. 382–399 (2023). Springer
- [9] Li, X., Li, Q.: Using bert and word definitions for implicit sentiment analysis. In: *2022 IEEE 24th Int Conf on High Performance Computing & Communications; 8th Int Conf on Data Science & Systems; 20th Int Conf on Smart City; 8th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys)*, pp. 1063–1068 (2022). IEEE
- [10] Wei, J., Liao, J., Yang, Z., Wang, S., Zhao, Q.: Bilstm with multi-polarity orthogonal attention for implicit sentiment analysis. *Neurocomputing* **383**, 165–173 (2020)
- [11] Wang, S., Zhou, J., Sun, C., Ye, J., Gui, T., Zhang, Q., Huang, X.: Causal intervention improves implicit sentiment analysis. *arXiv preprint arXiv:2208.09329* (2022)
- [12] Ouyang, J., Yang, Z., Liang, S., Wang, B., Wang, Y., Li, X.: Aspect-based sentiment analysis with explicit sentiment augmentations. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 18842–18850 (2024)
- [13] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial networks. *Communications of the ACM* **63**(11), 139–144 (2020)

- [14] Tran, N.-T., Tran, V.-H., Nguyen, B.-N., Yang, L., Cheung, N.-M.M.: Self-supervised gan: Analysis and improvement with multi-class minimax game. *Advances in neural information processing systems* **32** (2019)
- [15] Croce, D., Castellucci, G., Basili, R.: Gan-bert: Generative adversarial learning for robust text classification with a bunch of labeled examples. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 2114–2119 (2020)
- [16] Beltagy, I., Lo, K., Cohan, A.: Scibert: A pretrained language model for scientific text. *arXiv preprint arXiv:1903.10676* (2019)
- [17] Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C.H., Kang, J.: Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* **36**(4), 1234–1240 (2020)
- [18] Li, C., Xu, T., Zhu, J., Zhang, B.: Triple generative adversarial nets. *Advances in neural information processing systems* **30** (2017)
- [19] Guest, E., Vidgen, B., Mittos, A., Sastry, N., Tyson, G., Margetts, H.: An expert annotated dataset for the detection of online misogyny. In: *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pp. 1336–1350 (2021)
- [20] Mishra, S., Sturm, B.L., Dixon, S.: Local interpretable model-agnostic explanations for music content analysis. In: *ISMIR*, vol. 53, pp. 537–543 (2017)